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Proof by contradiction suppose we a chordal graph that does not have any vertices whose neighbours form a clique. Talked with Nate on this

Well that means we can't have any degree 1 vertices because it's neighbour would form a 1 clique. Thus every vertex must be at least degree 2 and that for every such vertex there is no edge between its neighbours. For vertices of larger degrees, at least one pair of its neighbours must not have an edge.

Thus suppose we take a longest induced path P from our graph G . Then look at its endpoint v . We know that v is at least degree 2 so it has another point i . Since i is not in P , it must connect back to P otherwise our longest path would include i . If i connects to the node adjacent to v then we are done, so we assume it doesn't. Thus means we would have a induced cycle of size greater than 3 because we can induce this path and our node i to create the cycle.

Thus v must have at least 2 neighbours i, j not in our path. From a similar argument, i, j must connect back to the path somewhere and if they don't connect back to the node adjacent to v then we can induce a cycle of size greater than 3 by removing the vertex adjacent to v that is not i or j . Suppose now that i and j are both connected to the vertex adjacent to v but not adjacent to each other. If they are only degree 2 then the neighbours of i or j produce a 2 clique (v and the unused neighbour). Thus i and j need to connect to some other vertex that is not in our path. This creates a contradiction because then we can induce a longer cycle by removing v and adding i and this new neighbour creating a new longest cycle but we assumed the original cycle was longest.

We repeat this argument for v being of degree k because we look for the one such pair of vertices adjacent to v that are not adjacent to each other and they would create a cycle of size greater than 4 or form a triangle in which case we remove it for our 2 clique.

Therefore chordal graphs have a vertex whose neighbours form a clique.

2

By Brook's theorem, the chromatic coloring of a complete graph is the number of vertices ($= \Delta(K) + 1$). Thus we can run our greedy algorithm starting with the nodes in our largest clique and then performing a BFS from the clique exploring the rest of the graph. As a reminder, the greedy algorithm always picks the smallest available color that is not used by any of its neighbours. Thus from Brook's theorem we know that we will use at least $\omega(G)$ colors but suppose that we end up using more than that in our algorithm (assume $\omega(G) + 1$ for now). That would imply we have a vertex that is connected to all points in a clique. That would create a contradiction because then our largest clique plus this vertex would form an even larger clique.

Therefore we can color our graph with $\omega(G)$ colors and we know that this coloring is optimal because by Brook's theorem our clique subgraph requires at least $\omega(G)$ colors